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A Conversational Assistant for Democratization of Data Visualization: A Comparative Study of Two Approaches of Interaction

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ABSTRACT

Data visualization is an important aspect of exploratory data analysis which requires expertise, posing a challenge to many users. In this work, we present a conversational system designed to make data visualization accessible, where users can generate visualizations of data by natural language conversational interaction with the system. We conducted a between-subject study to compare two versions of our system, where users were asked to perform data exploration tasks. We did not find significant differences as concerns user assessments of their interactions with the system; however, we uncovered that users explore different properties of the data and reach different conclusions when using the two systems. We will discuss the general implications of our findings as regards the usage of conversational interfaces for visualization.

1 | Introduction

Visualization systems are powerful systems that help users explore their ever-growing datasets, and enable new discoveries. However, a key challenge hindering users from adopting the latest visualization tools and techniques is the steep learning curve that has to be overcome in order to make use of them. Users would like to be able to simply tell the visualization system what they want to do: namely, they would like to engage with a conversational assistant (CA), which should function as a mediator between the user and the visualization system.

It is at least 20 years that several research groups, including ours, have engaged in research in the space of CAs for visualization [1]. Our specific goals in this larger research space are to develop natural language interfaces (NLIs) for users to produce visualizations rendered on *large* screen displays—the latter have been shown to better assist users in data exploration, especially when users collaborate on a task [2–4].

In this paper, we have developed two different CA architectures and discuss the user study conducted with them. Importantly, both CAs are "always-listening," namely, they do not act only when directly addressed with a command, as happened in many NLIs; rather, they actively listen, respond to implicit commands, and use the context for interpretation. In our studies, two users work together as a team to explore the data, and hence the CA

Abbreviations: DM, Dialogue Manager, dialogue manager; NLI, Natural language Interface; NLP, Natural Language processing; PE, Property Extractor Architecture.

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listens in to the conversation between the two, and intervenes appropriately, similarly to what another team member would do.

The two NLI engines differ in their algorithmic approach to understanding the users, which potentially results in different effectiveness of the interface, and a different experience for the users. In this paper, we strive to evaluate a variety of important characteristics of the two NLIs in order to understand whether one of them is preferable to the other, or, as we will conclude, whether a hybrid of the two would be the better choice.

The first aspect we will examine is the *functionality* itself of the two NLIs. Since the CA is an interactive system, the charts produced after processing the input utterances, and their types, are the artifacts of the interaction between the users and the system. Given two different NLIs that behave potentially differently, the users' behavior may change as well, in a feedback loop between system and user behaviors. This interaction can be quantified in terms of number of utterances processed, number and types of charts generated as well as the database properties explored by the users during the studies. Next, the CA system should be *interpretable*: for a system to be interpretable, the user should be able to understand its output [5]. Being a latent quality, oftentimes interpretability is difficult to measure [6]; hence, we direct our attention to post hoc interpretability which pertains to information obtained after a model has been deployed [7]. Finally, we look at the *usability* of the system, since it has long been established as one of the indispensable qualities of a NLI [8–10].

We articulate the three features we discussed above, namely, *Functionality*, *Interpretability*, and *Usability* through the following three research questions:

- **RQ1 (*Functionality*):** Do changes in the NLP engine affect user behavior, and if yes, how?
- **RQ2 (*Interpretability*):** Do changes in the NLP engine affect the conclusions that users draw from their explorations, and if yes, how?
- **RQ3 (*Usability*):** Do changes in the NLP engine affect users' perceptions of the NLI?

These RQs help us reflect on the behaviors of the two NLP engines focusing on the user's point of view. Whereas we will find that the answers to majority of the RQ's are "yes," we will also find that for RQ1, these changes affect a small set of variables, namely, the number of processed utterances, the number of generated charts of some specific types, and importantly, the properties explored by the users; they affect the kind of conclusions that the users derive (RQ2), raising a broader issue concerning the quality of the guidance these systems offer to the users; they do not affect the users' perceptions of the ease of use, or usefulness of the system (RQ3).

The paper is structured as follows. After describing related work in Section 2, we provide background to our work in Section 3, since the NLI architectures we present here are grounded in previous research of our group. Section 4 presents the two different NLI architectures, while Section 5 describes the user study we conducted. Section 6 provides a detailed analysis of the differences between the two systems as concerns the three research questions above. Finally, the takeaways from the study are discussed in Section 7, and Section 8 presents several avenues for future work.

2 | Related Work

Earlier work on CAs for data visualization include the work by Cox et al. [11], which was perhaps the first to establish the benefits of using NL to generate visualizations for exploratory data analysis. In the ensuing 20 years, several NLIs were designed to support free-form NL queries as input for generating visualizations. Shen et al. [1] present a systematic survey based on different aspects like the NL pipeline, system design, designing interactions with users and so on. Among the systems from the period 2015–2020, we highlight a few which opened additional venues for investigation, and/or provided further evidence that users benefit from such NLIs. DataTone [12] used language, but when it was not able to resolve an ambiguity, it presented the user with "selection widgets," but no further conversation. Eviza [13] provided a dialogue interface for users to explore existing visualizations, but did not support creating a new visualization or modifying an existing one. These limitations were addressed in Eviza's successor Evizeon [14], which focused on users typing in follow-up queries, however, did not use speech or gestures. Evizeon deserves credit though, for providing further evidence that "natural language interaction can offer numerous advantages in terms of ease of use, convenience, and accessibility to novice users, facilitating the flow of analysis for novices and experts alike" [14]. FlowSense [15] allowed domain experts to type queries or use speech through a web front end to construct dataflow diagrams; Voder [16] focused on augmenting visualizations with automatically generated facts; NL4DV [17] was conceived as a python API to be easily integrated into an existing visualization system pipeline, but it did not handle follow-up queries. Other work investigated related facets of the problems, for example, creating graphics from text [18], or editing graphics via an NLI [19], but did not aim to use NLIs support a full-fledged spoken conversation with the user.

More recently, DT2VIZ [20] focused on a deeper understanding of a user query in order to explore the data, but the queries were typed, most often in the form of keywords that the system autocompleted, and the system itself suggested follow-up queries. ncNet [21] is most likely the first end-to-end NLI for visualization, and maps directly from a language query to a visualization; its architecture is based on transformers. It is trained on the nvBench dataset, that includes 25,750 NL queries and the corresponding visualizations, that is, 25,750 (NL, VIS) pairs; however, it is not trained on conversations on data explorations that lead to visualizations.

Lately, large language models (LLMs) have started being integrated into visualization tools (e.g., PandasAI from OpenAI[1]), but not so far as part of a CA that keeps track of dialogue history (DH). Chat2Viz [22] does explore using three different LLMs to generate visualizations from natural language with promising results; however, the approach is not interactive and makes use of

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complex prompts, which detract from the naturalness of a conversation. Additionally, Chat2Viz is evaluated on the nVBench dataset, and not with user study; indeed, we are not aware of any NLI's for visualization grounded in LLMs that have been formally evaluated with user study.

3 | Background: Contextualizing This Study

It is more than 10 years that we have been engaged in a quest for a completely natural CA to support exploratory data visualization on large screens. Over the years, the NLIs we have designed and deployed have been endowed with an expanding repertoire of interaction modalities, and with increasingly sophisticated speech recognition (SR) and natural language understanding (NLU) capabilities. The first pilot prototype we developed [23, 24] produced statistical visualizations (bar charts, scatterplots, histograms, etc.) from simple commands. In this prototype, a spoken query was recognized by the Chant Speech Kit,[2] and then mapped via the Stanford Parser [25, 26] into a nine-feature vector. These visualization features were learnt via supervised learning (C4.5 [27]) over a corpus of sample queries, given the chart classification model in [28]. We compared our prototype against popular graphing tools of the time, such as Microsoft Excel or Tableau. The findings were highly encouraging. At least half of the graphing-tools users had to create multiple charts to describe a query that could have been expressed with a single sentence; and on average users took 12 times longer to realize a graph for a query in the graphing-tool than in our prototype. Users remarked that the language-based system allowed them to focus more of their attention on deriving hypotheses about the data rather than the mechanics of producing the right visualization to prove their hypotheses.

While this result demonstrated that natural language would be a viable way of generating visualizations, the queries were not part of a conversation; indeed, we realized we did not know how a user would actually interact with a CA to explore visualizations. Hence, we undertook a data collection, which resulted in a corpus of multimodal interactions for 16 subjects that explored public crime data in our city to better deploy police officers.[3] Those users spoke with a human visualization executor (VE) who remotely created visualizations on a large screen, was not visible and did not speak back. The users were asked to reason out loud about their understanding of the exploratory task discovered through visualization. On the basis of that corpus, which is publicly available,[4] we developed Articulate-2 [29–31], based on a relatively standard dialogue system pipeline with distinct modules for speech recognition, syntactic parsing resulting in a meaning representation, and dialogue management including dialogue act (intent) recognition. When appropriate, the NLP engine transforms the user request into an SQL query, and in a visualization specification that is passed to Vega-lite2,[5] a separate visualization interface software, which creates the visualization of the data returned by the SQL query and adds it to the display. A first version of that NLU interface was evaluated with 18 subjects [30]. Subjects were given sets of predefined questions to ask (only typed, only spoken, spoken + gestures), and finally invited to ask a few questions of their own choosing. Surprisingly to us, on several usability questions, the system fared significantly better when the users could ask their own questions notwithstanding fewer successful visualizations created. This shows that users are satisfied with a system that allows them to interact freely, even if it makes mistakes, as opposed to obtaining more success via preselected queries.

Building on this body of work, most recently we have designed an architecture of an always-listening CA, with which users can interact freely, and which is the focus of this paper. In contrast to our previous work, in which a single user was directly interacting with the assistant, in this version two users collaborate on a data exploration task, hence bringing the design closer to supporting real-world exploration of data by teams of users (see Figure 1). The assistant functions as a third participant (moving fluidly between the roles of a direct interlocutor and an overhearer), which constantly listens to the conversations: this helps it to understand current context and the requests made by the users. In what follows, we will first describe the always-listening NLI comprehensively, but then delve into the two different NLP engines, and the user study that compares the two, which is the main topic of this paper. We have conducted user study with this always-listening architecture [33, 34] and found that:

- The NLI responds to direct requests from users as well as indirect queries, as a result of the always-listening feature.
- The always-listening model helps to leverage additional context obtained by overhearing the users when they talk to themselves or think out aloud. This results in useful charts being generated without the participants' explicit request.
- The CA's proactive nature in generating charts helps to reduce the cognitive overload of interacting with a new interface.
- It is also capable of dealing with incorrectly formed queries and generating reasonable charts. These incorrect queries either result from speech-to-text error and/or from the natural flow of human conversations, which are full of pauses, restarts, fragments, and occasional overlap between speakers.
- Overall, the CA is capable of catering to a multi-user collaborative environment and generate visualizations of data to aid exploratory data analysis.

4 | Always-Listening NLI Architecture

The always-listening NLI architecture (see Figure 2) has *five* major components. First, the *Always-listening speech-to-text* module enables the system to continuously listen to the users' conversation. In our earlier work [34], we found that a constantly listening interface being capable of "overhearing" the users can pick up useful context from the conversation and respond with visualizations irrespective of whether it is directly addressed to or not. This helps the users converse freely without the need to pause and say wake words (like "Hey Siri!," "OK Google!," or "Alexa!") to make the system listen to their requests. We demonstrate this later in the paper using examples of conversations and generated visualizations from the user study in Figure 8. To implement this

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FIGURE 1 | A snapshot from one of the groups of the user study where two participants are seen to interact with the articulate NLI [32].

always-listening speech-to-text module, we use the Web Speech API[6] as done in our earlier work [33, 34].

The next component in the architecture is the *NLP engine* which processes the textual interpretation of the spoken utterance[7] on the part of the speech-to-text module. The NLP engine is responsible for processing each utterance and determining if it is a request or not. It also extracts the entities from the utterance relevant to the data being explored and generates the corresponding chart metadata. In this paper, our goal is to compare the performances of two NLP engines, elaborated in Sections 4.1 and 4.2.

The chart metadata generated by the NLP engine contains Vega-Lite grammar encodings of the visualization to be generated (see Section 4.2.2). The Vega-Lite encodings are sent to the *Visualization Generator* module where the chart graphics are produced and rendered on a "conveyor belt" on the display screen (see Figure 2, Label 6), which is the final output of the NLI architecture. The interface enables the user to select, hover upon to magnify, move, resize, and close the generated charts and use them in the assigned exploratory data analysis tasks. The current system supports the following chart types—choropleth maps, heatmaps, bar charts, line charts, and histograms; Figure 3 illustrates these different charts and provides a definition and an example for each. Please note that the Property Extractor Architecture generated histograms during the user study. However, no histograms were generated during the Dialogue Manager Architecture user study, so in this comparative analysis.

We now turn to describe the two different NLP engines.

4.1 | Property Extractor Architecture

Identification of database properties (database attributes or filters) whether it is explicitly stated or implied in the user's utterance is the core function of this NLP engine [34] (see Figure 4a). When an input utterance reaches this module, it determines whether it is a request for a chart or not using one of two methods—*Chart Extractor*, when a query has an explicit mention of any of the chart types currently supported by the system, determined with keyword match; *Chart Classifier Neural Network*, which was trained using the NLP.js library[8] on a small dataset of 1372 utterances collected during a pilot evaluation with 4 participants (2 groups) and which achieved a 95.19% accuracy on the test data (60 − 20 − 20 training–validation–test split) [33].

Once a chart type is identified using either of the methods, the NLP engine needs to extract one, two or three database properties or attributes from the utterance for generating that chart type. Since some kinds of charts work better with certain types of data, the method suggested by NL4DV [17] is used, which lists a series of mappings linking data types to chart types. These mappings are listed in Figure 5. These database properties are extracted using one of the two competing methods—*Explicit Property Extractor* and *Implicit Property Extractor*. As the name suggests the former detects property names explicitly mentioned in the utterance. The *Implicit Property Extractor* uses a predefined sliding window of utterances (a window of five utterances is used in the study) to obtain context for the current utterance and a threshold on the number of times any database properties are mentioned in the sliding window to be extracted. Using these additional properties from the context as well as those from the current utterance,

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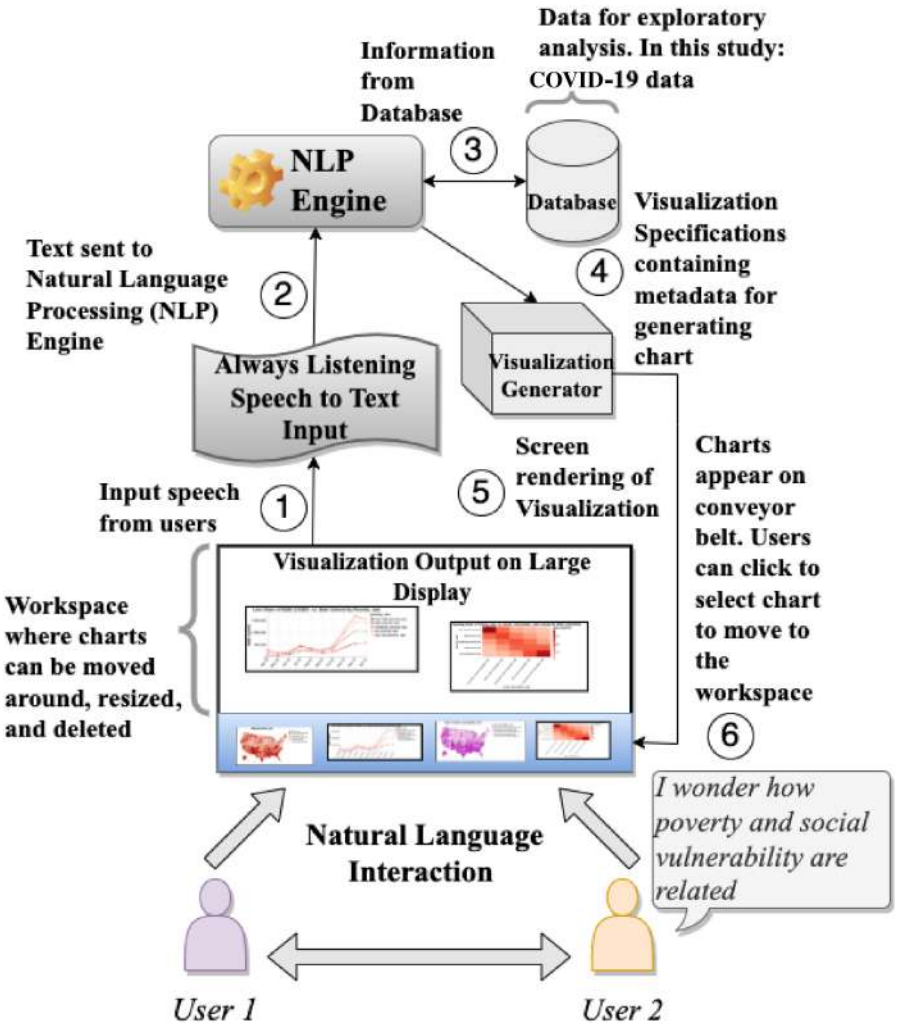


FIGURE 2 | Always-listening NLI architecture [32].

chart metadata is generated. Thus, the *Implicit Property Extractor* leverages the always-listening feature of the architecture to ascertain the database properties for the visualization generation. Additionally, at a timed interval of 10 min, spontaneous charts are generated called Intermittent Spontaneously Generated Charts where the system acts as a proactive agent to help the participants in their exploratory tasks. Here, the chart specifications are determined based on the properties extracted from the current sliding window. Once a chart type is determined and database properties are extracted, the information is sent to the visualization generation module for appropriate specification generation for the chart.[9] The Property Extractor Architecture (PE) version of the CA underwent a preliminary evaluation with 14 participants (7 groups) that focused on qualitative high-level properties, like assessing that it can react to both direct and indirect queries [34]. However, in the study we report here, the PE NLP engine was directly compared to the DM NLP engine in exactly the same environment; we describe the DM next.

4.2 | Dialogue Manager Architecture

The Dialogue Manager Architecture[10] is adapted for the current always-listening application from Articulate-2, the earlier version of the CA [30] that we briefly described above. It is a full-fledged dialogue system with NLU components including Semantic Slot Filler grounded in a Knowledge Ontology (KO), dialogue act classifier, and reference resolution module. A Dialogue Manager (DM) component maintains the DH and the generated chart metadata. These components are described below.

4.2.1 | NLU Components

The NLU module consists of the following components:

- *Knowledge Ontology*:

A computational ontology can be defined as a way of modeling the structure knowledge of a system using relevant entities and relations—as a generalization/specialization hierarchy [35]. The terms "generalization" and "specialization" are used to explain an entity type and value of the entity type in the computational KO, respectively. The KO is created with information from the database, hence the "generalization" is analogous to the "attribute name" or "column name" of the database. The specialization is the "attribute value" or "filter" in the database. For the tasks in the user study, we used COVID-19

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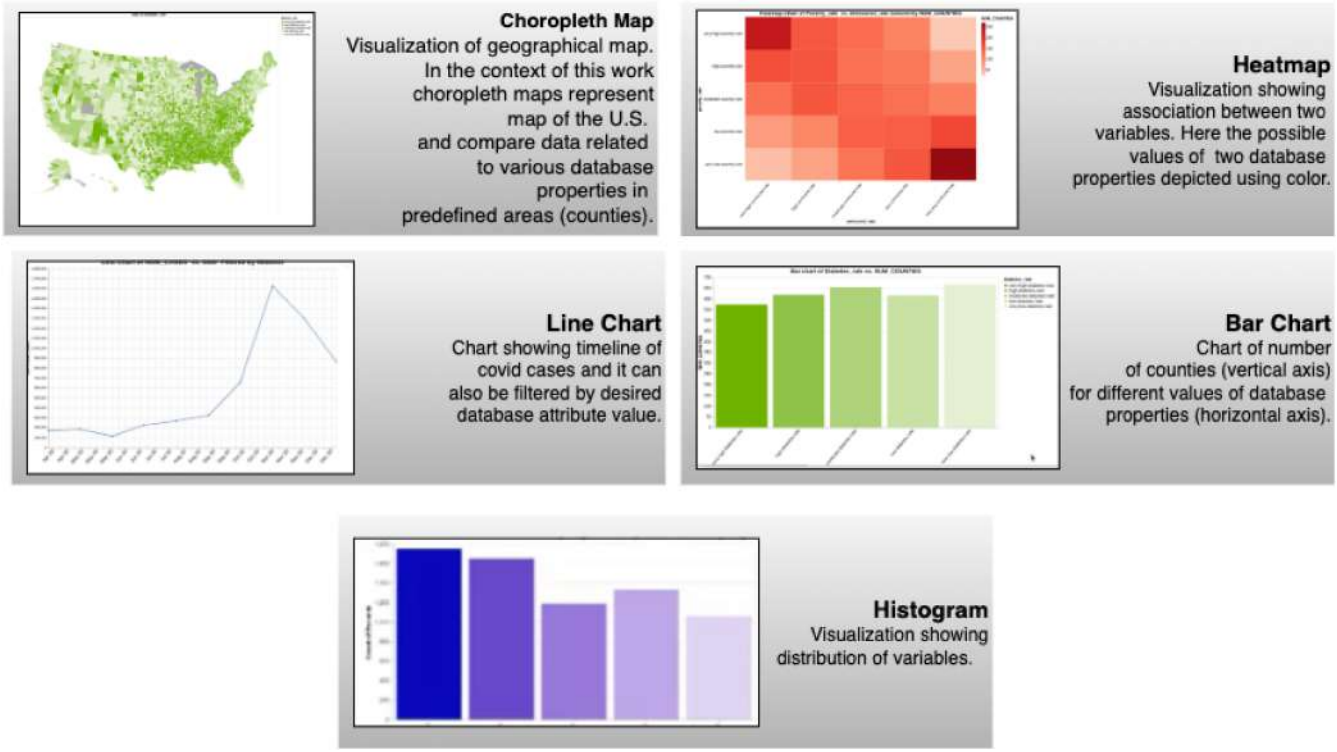


FIGURE 3 | Chart types generated during the user study.

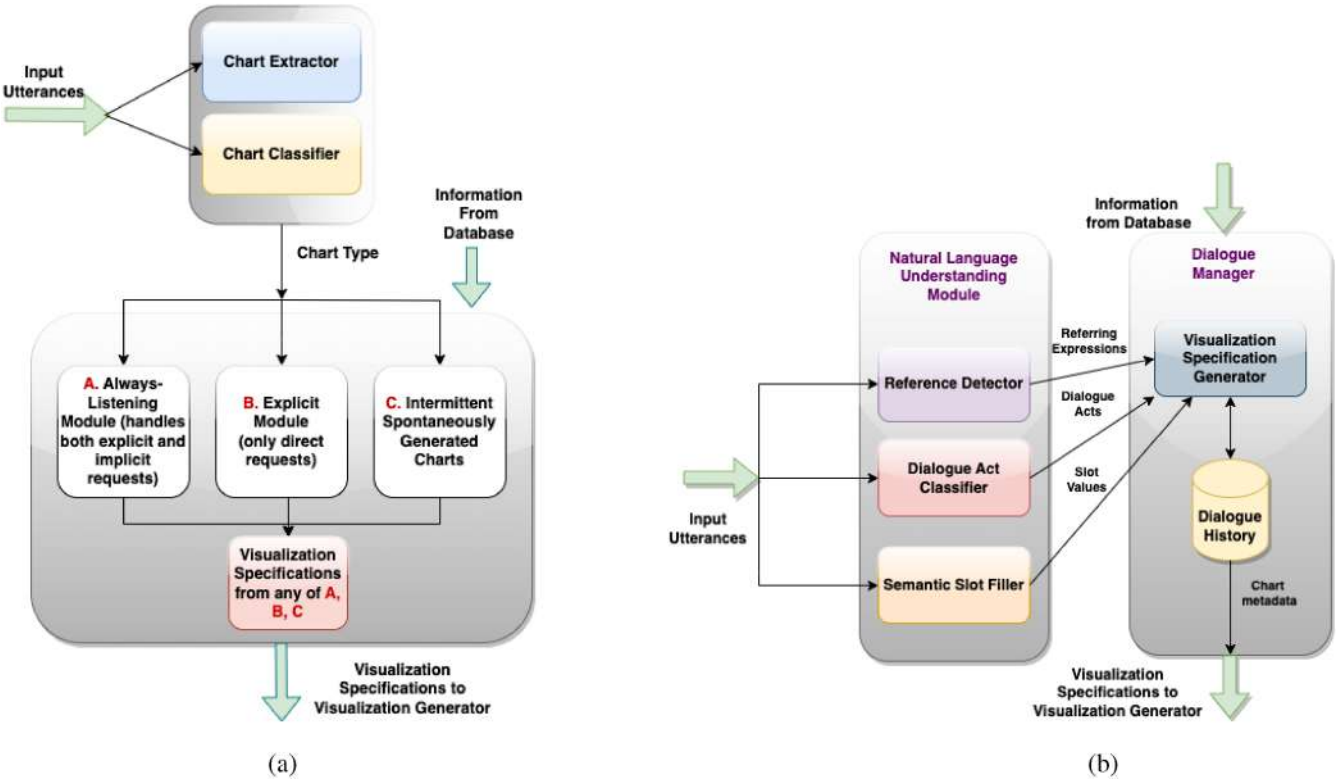


FIGURE 4 | The NLP engines. (a) Property Extractor Architecture. (b) Dialogue Manager Architecture.

data [36]—a dataset for each county in the United States which includes the CDC's (Centers for Disease Control and Prevention) social and healthcare factors like *diabetes rate*, *access to doctors*, *uninsured rate*, *social vulnerability rate*, and so forth and the number of COVID-19 cases from April 2020 to January 2021 (see Figure 8). Based on this COVID-19 dataset and following the works of Kumar et al. [30] and Kumar [37], we created the KO semi automatically using synsets (sets of synonyms)

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Chart Type	Data (Essential)	Data (Optional)	Example
Choropleth Map	Nominal Attribute × FIPS (Federal Information Processing System) Code	Nominal Attribute Value	Choropleth Map of Diabetes Risk (Filtered by Very High Covid Vulnerability Rank)
Heatmap	Nominal Attribute (*X*-Axis) × Nominal Attribute (*Y*- Axis)	Nominal Attribute Value	Heatmap of Poverty Rate and Uninsured Rate (Filtered by Southeast and Southwest)
Bar Chart	Nominal Attribute (*X*-Axis) × Quantitative Attribute (*Y*-Axis)	Nominal Attribute Value	Bar Chart of Diabetes Risk vs Number of Counties (Filtered by Midwest)
Line Chart	Temporal Attribute(*X*-Axis) × Quantitative Attribute (*Y*-Axis)	Nominal Attribute Value	Line Chart of COVID Cases over Time Span of 10 months (Filtered by Very Low Cardiovascular Disease Rate)
Histogram	Nominal Attribute (*X*-Axis)	—	Histogram of Uninsured Rate

FIGURE 5 | Mappings linking data types to chart types. In the examples, the optional nominal data, highlighted in blue is used as the filtering criterion [34] (for FIPS code, see https://transition.fcc.gov/oet/info/maps/census/fips/fips.txt).

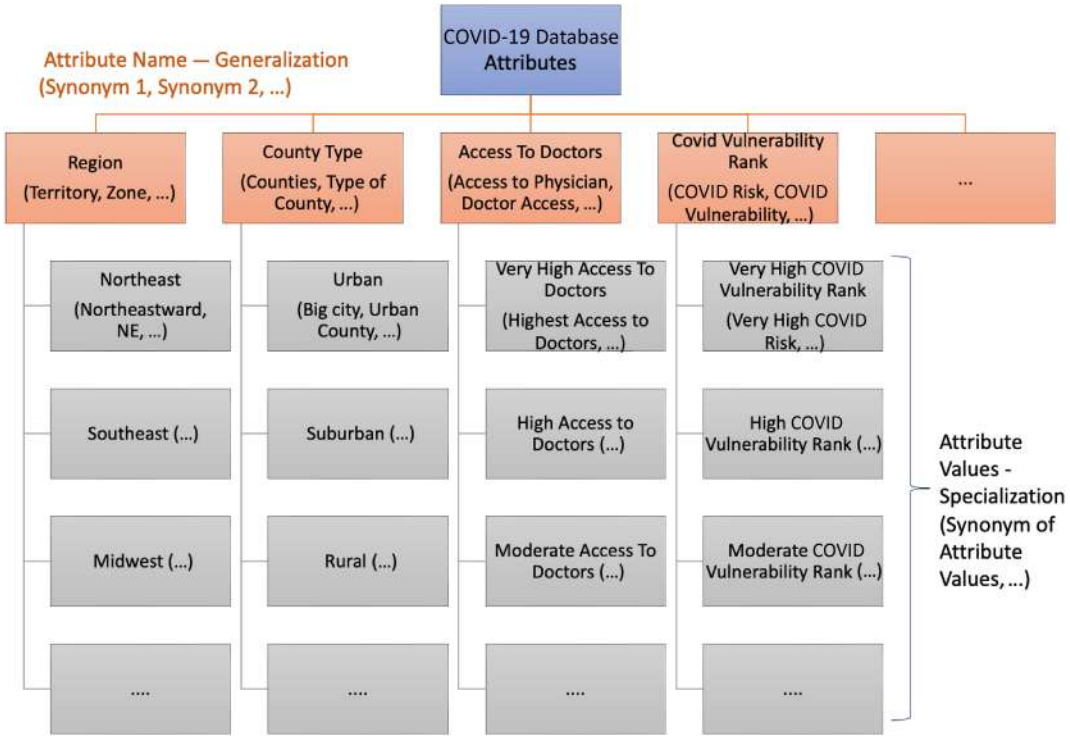


FIGURE 6 | Structure of the Knowledge Ontology.

extracted from Wordnet (https://wordnet.princeton.edu/) and Babelnet (https://babelnet.org/).

An example of the KO for this work can be seen in Figure 6. Here, the generalization *Region* has seven possible values—*Northeast*, *Midwest*, *Southeast*, *Southwest*, *Rockies*, *Pacific*, and *Non-contiguous*, each being a specialization. Synonyms generated for *Midwest* were *American midwest*, *Midwestern United states*, and so on. Overall, there were 13 generalizations and a total generated vocabulary of 790 terms with specializations and their synonyms. The KO forms the foundation of Semantic Slot Filling.

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- *Semantic Slot Filling*: Semantic slots are those tokens in an utterance which are predefined by the KO. The semantic slot filling component is thus responsible for identifying tokens (words or phrases) in an input utterance belonging to predetermined semantic slots and replacing those with the *slot values* as defined by the KO. For example, in the utterance "Can we see COVID risk in the midwest?" the tokens *COVID risk* and *midwest* are identified and matched to the predetermined semantic slots *COVID–vulnerability–rank* and *Region*, respectively, from the KO. Following our previous work [38], we adopted a two-phase approach for word-level and phrase-level semantic slot identification. In the first phase each utterance is tokenized into words using the NLTK Tokenizer Package,[11] following which some tokens are merged to form longer tokens based on KO matches (*COVID risk* is considered one token instead of two). The tokens which are classified as nouns (namely, have a Noun part-of-speech tag) are matched with the KO terms and assigned semantic slots. Each utterance u and suggested semantic slot s from the first phase are passed on to the next phase. Here, as appropriate, tokens are checked for compound nouns (e.g., "area" and "type" concatenated to form "area type"), prepositional phrases (p) are interpreted ("area type in the United States") and semantic slots are assigned ("area type in the United States" assigned to the semantic slot "county-type"). Following this two-phase algorithm, the utterance with identified slots ("Can I see [COVID risk, COVID-vulnerability-rank] by [area type in the United States, county-type]?") is delexicalized. Delexicalization involves replacing the spans of text identified in the previous step by the semantic slot names ("Can I see [COVID-vulnerability-rank] by [county-type]?").
- *Dialogue Act Classification and Request Identification*: After semantic slot filling, the intent of the utterance is determined through the Dialogue Act Classifier (DAC). Dialogue acts, which were originally based on speech act theory from linguistics [39, 40], capture the speaker's intent from an utterance. Many different inventories of dialogue acts exist [41, 42] that capture different dimensions of communication (such as asserting new information, or providing feedback to the interlocutor as in an answer, or managing turn allocation). In our current NLI, we use the same inventory of eight domain- and task-specific dialogue acts that we developed on the basis of the user study on crime in Chicago we mentioned earlier [29, 30]. These dialogue acts are: Create Visualization, Modify Visualization, Clarification, Window Management, Fact Based, Preference Based, Appearance, and High-Level Query. Using general inventories of dialogue acts, while theoretically warranted, would have been too complex for our purposes and small corpus; hence, we developed this set inductively—bottom-up, on the basis of the transcribed data—and validated it as follows. In this user study, 16 subjects explored public crime data in the city of Chicago. The users spoke with a human VE who remotely created visualizations on a large screen, was not visible, and did not speak back. The users were asked to reason out loud about their understanding of the exploratory task discovered through visualization.

In the crime corpus, all 3179 utterances were transcribed; coders were then asked to identify actionable requests (ARs), namely, those utterances that directly affected what the human VE was doing ($k = 0.84$ on this binary task). Subsequently, the 449 resulting ARs were labeled for the dialogue acts listed above, and in this 8-way annotation process, an excellent inter-coder agreement ($k = 0.74$) was obtained. The utterances assigned with DAs CREATE-VIS or MODIFY-VIS identify the ARs that are requests to the system. CREATE-VIS is the DA for creating a new chart based on the data attributes found in the user's request, while MODIFY-VIS signifies generating a new chart on the basis of an already existing chart (used during reference resolution).

Additionally, we observed that the AR was often preceded by context that was needed for its interpretation, especially for reference resolution. We then defined CARs. As its name suggests, a CAR is an AR in a given context defined by—(i) a *setup* consisting of one or more think aloud utterances, up to and including an utterance that mentions data attributes, if any, (ii) an AR and (iii) finally one or more utterances following the AR. The context provided by the CAR is used for reference resolution, to which we turn now.

Reference Detection and Resolution. The NLU module also performs reference detection and resolution (further details on our reference detection and resolution pipeline can be found in the work by Bhattacharya et al. [32]). In reference resolution, references to entities previously mentioned or in our setting, previously generated during the interaction are detected and resolved—for example, in "Can I see *the same chart* for the midwest," the reference *the same chart* needs to be matched to an existing chart, certain properties of which are retained (e.g., its type), but others are changed (e.g., from *southwest* to *midwest*). The CARs collected from the user study exploring city crime were labeled for text references (with excellent inter-coder agreement of $\kappa = 0.85$ [43]) and 176 text references were obtained—the analysis on the Chicago-crime-viz corpus was limited to the CARs, since our interest was in resolving references within an AR. A sequence tagging CRF model was trained with parts-of-speech tags as features where the model predicts tags using the standard IOB2 format (B-REF for Beginning, I-REF for Inside and O-REF for Outside text references in a given utterance). For example, in the utterance "Can I see *the same chart* for the midwest," *the* would be labeled with B-REF and *same*, *chart* as I-REF while rest of the words (tokens) would be labeled as O-REF.

References are resolved for those utterances classified as MODIFY-VIS by the DA classifier having a detected reference with the help of heuristics based on recency metric and cosine similarity. Given the DH (described later in Section 4.2.2) with n utterances, the word embedding vectors of the most recent $n/2$ utterances (w.r.t. the current utterance) are matched with word embedding vectors of earlier utterances stored in the Dialogue State History using cosine similarity. The utterance in the DH having the highest score is selected as the best match for resolving references. All these metadata from the NLU component—the DA, Semantic Slot values, the detected reference, and referent information, if any—are sent to the DM.

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4.2.2 | DM Components

The DM module is responsible for taking information from the NLU module and generating visualization specifications for the charts with this information using the Visualization Specification Generator (VSG). It also adds these inputs and the visualization specifications to the DH. These two components are described below.

- *VSG*: Information from the NLU module—the identified semantic slot values, the dialogue acts, and any information of detected reference, are the inputs to the VSG. Based on the semantic slot values and heuristics, chart metadata are determined. Chart metadata includes type of chart (bar chart, heat map, etc.), horizontal and vertical axes (if applicable) and the database properties found in the semantic slots of the utterance. Chart types and the axes are determined using the mappings described in Figure 5. The VSG is also responsible for generating the SQL query for the current utterance based on a set of predefined rules derived from the aforementioned mappings between data types and visualizations. The generated SQL query is then sent to the database and the fetched data are transformed into a specification expressed via the Vega-Lite grammar. These are JSON encodings for describing charts which, when interpreted by a Vega-Lite compiler, generate the actual visualization of the data. The Vega-Lite specifications along with the previously mentioned chart metadata are sent to the Visualization Generator as well as the Dialogue State History in the form of a JSON object.
- *DH*: This component stores the entire conversation in the order of the utterances spoken by the users while interacting with the system. It also stores the dialogue act, detected references and visualization specifications if an utterance results in chart generation. This DH is used extensively for resolving references.

5 | User Study

For ease of understanding, we will refer to the user study with the Property Extractor NLP engine as *PE* and the one with the DM NLP engine as *DM*. The setup of the studies with both PE and DM is the same.

5.1 | Setup

The participants for both studies were adults (age > 18) recruited at the University of Illinois Chicago. For PE we recruited 20 participants (10 groups of 2) and for DM we recruited 30 participants (15 groups of 2). All 50 participants were compensated for their time. The studies for PE and DM were done separately (PE first and DM 2 months later), and we ensured there was no overlapping participation. The entire study session lasted about an hour including the 50 min for the two tasks (25 min each). Each participant was provided with a task sheet including the two task questions, a list of database properties (attributes and filters) in the COVID-19 dataset [36] for easy reference, a mouse for moving, resizing, hovering, and selecting the generated charts (even though the participants had a mouse each they still had to share control of the mouse pointer on the interface), a pen and a paper for notes. The utterances of the participants were fed to the NLI through ceiling microphones and charts appeared on the "conveyor belt" of the interface (as described in Section 4). They interacted with the CA by on a large-screen display as seen in Figure 1. Here, we see a setup from one of the user study sessions where the participants can be seen performing one of the exploratory tasks using the NLI. An overview of the COVID-19 dataset explored by the participants is shown in Figure 7.[12] The attributes from *Elderly Population* to *COVID Vulnerability Rank* in Figure 7 can have five possible values—*very low*, *low*, *moderate*, *high*, and *very high*. For example, *Uninsured Rate* has the following values—*very low uninsured rate*, *low uninsured rate*, *moderate uninsured rate*, *high uninsured rate*, and *very high uninsured rate*.

5.2 | Tasks

For the exploratory data analysis tasks, the participants were provided with a prompt and a set of two *open ended* exploratory tasks. We choose open-ended tasks so that we are not constrained by whether the users reach a correct conclusions, but rather focus on how the participants explore the data in order to answer the three research questions introduced earlier. These tasks were used in the preliminary evaluation of the PE version of our NLI [33, 34].

"*Arti is a virtual assistant who is always-listening and will be making visualizations for you. You will be discussing with each other, including Arti, to explore COVID-19 data. Your task is to create and select visualizations that Arti creates that you think will be useful to answer the 2 questions in your task list. You will be given 25 min for each task and at the end of 25 min, we will listen to your answer and refer to us which of the visualizations helped you get to that answer.*

1. *We are looking for input from US residents about how to prioritize resources, based on county-level data. Essentially, there are different ways to group US counties based on region, the county type (e.g., rural, urban, suburban, small city), and other county-level data, in order to best reach highly at-risk communities. What regions and area types in the US should we prioritize for directing resources based on*:

   a. *Covid risk*
   b. *Poverty*
   c. *Diabetes*
   d. *Cardiovascular disease*
   e. *Elderly population*
   f. *Uninsured*
   g. *Access to doctors*

   You will be creating visualizations that will be used as evidence to support your claims that answer the question above or any conclusions that you made based on the data.

2. *Are there interesting patterns within different sets of counties (based on different combinations of parameters) that might help us understand COVID-19 outbreaks and better prioritize resources?*

   Example combinations of parameters:

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Attribute	Description
FIPS Code	U.S. county-wise FIPS codes.
Region	Each county can belong to one of the areas in the United States — Northeast, Midwest, Southwest, etc.
County type	Urban, Suburban, Rural, Small city
Elderly Percentage	Percentage of people aged 65 and over in each county
Access to Doctors	The opportunity or right to receive health care by county
Uninsured Rate	Uninsured population in each county
Diabetes Rate	Rate of Diabetes risk in each county
Cardiovascular Disease Rate	Rate of Cardiovascular disease risk in each county
Poverty Rate	County-wise poverty rate
African American Population	African American population in each county
Hispanic Population	Hispanic population in each county
Social Vulnerability Rank	County-wise social vulnerability
Covid Vulnerability Rank	County-wise risk of contracting COVID-19
Dates	Dates for collecting data regarding the number of COVID cases for each county. Total 10 dates, one for each month from April 2020 to January 2021
COVID Cases	Number of cases of COVID-19 collected on 10 dates, one for each month from April 2020 to January 2021 for each county

FIGURE 7 | COVID-19 dataset [36] used for the exploratory tasks in the user study.

a. *Midwest + rural + very high poverty rate*

b. *Southwest + suburban + very high elderly population + very low access to doctors*

You will be creating visualizations that will be used as evidence to support your claims that answer the question above or any conclusions that you made based on the data."

The first task is open ended where the users are encouraged to explore the breadth of the dataset by generating charts related to some given database attributes. Our objective was also to give them an opportunity to play with the NLI and get comfortable generating charts by interacting with the CA. The second task on the other hand is more specific, which asks the users to combine different database attributes and come up with a combination which they think is an interesting pattern in order to better prioritize resources in the occasion of a COVID-19 outbreak. The participants were not given any explicit instruction on how to use the system and were left to explore it by themselves. This was done to understand the ease of the NLI system use [44]. They were also encouraged to discuss the tasks among themselves and think aloud so as to ensure capturing as much context as possible from the conversations. They were also informed that the task questions are open ended and do not have a correct answer, so that they could explore the data without any pressure to determine the correct answers. At the end of each task they were asked to share their answers verbally. The entire procedure was audio and video recorded after obtaining written consent of the participants. Brief excerpts of conversations with the NLI alongside charts produced by the system from two groups of participants — one from PE and the other from DM — can be found in Figure 8a,b, respectively. From these conversations, we can observe that none of the utterances that generate charts are direct commands to the system; even more remarkable, only one of them specifically states the type of chart, a map (see *Should we get a map of the county type first* in Figure 8b); and one doesn't even include a verb (see *Times*, *city*, *rural*, *urban and the number of cases* in Figure 8a).

6 | Analysis of User Behavior, Inferences, and Perceptions

In Section 1, we introduced the three features of the CA — *User-System Interactivity*, *Interpretability*, and *Usability* around which we designed our comparative study of the two NLP engines. In this section, we elaborate on each of these features to support our overall hypotheses.

The number of utterances processed by the system, the number and types of charts produced during the exploratory tasks are artifacts of the system behavior as well as the user behavior; namely, the changes in user behavior and system behavior with different NLP engines go hand in hand, as can be seen from the examples presented in Figure 8. In turn, these differences may affect the way users navigate the tasks (in terms of the database attributes they explore), the conclusions the users reach at the end of each task, and their perceptions of the NLIs. We articulated these hypotheses as the following three research questions, introduced in Section 1 and repeated here for convenience:

- **RQ1 (*Functionality*):** Do changes in the NLP engine affect user behavior, and if yes, how?
- **RQ2 (*Interpretability*):** Do changes in the NLP engine affect the conclusions that users draw from their explorations, and if yes, how?

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(a)

Conversation	Chart Generated
U1: I'm just trying to make combinations.	—
U2: So, like places with high elderly percentage and probably the people that are going to have the most deaths. Table has to be compared with the tables. Or a particular region can get the data like it's higher or lower. Instead of using a map.	—
U1: Yeah, that's the only region that's higher. Okay.	—
U1: How do you do more than two?	—
U2: Okay. Interesting	—
U1: You can derive the problems for each county	Bar Chart of County type versus Number of Covid Cases
U2: What? What was that?	—
U1: Times, city, rural, urban and the number of cases.	Line Chart of COVID Cases versus Date, Filtered by Rural and Urban
U1: It's the number of cases	—

(b)

Conversation	Chart Generated
U1: I think we can do COVID risk first, right?	Choropleth Map by COVID Vulnerability Rank
U2: We have a very high COVID rank.	—
U2: What do you think about it?	—
U1: Should we get a map of the county type first?	Choropleth Map by County type
U1: Actually, it doesn't say that, well, what is the ongoing problem right now in America?	—
U2: The first is poverty, right?	—
U2: So, I think, wait.	—
U1: Well, I think we can do like COVID risk and then access to doctors and then...	Heat Map by Access to Doctor, COVID Vulnerability Rank

FIGURE 8 | Brief excerpts of dialogue with the Natural Language Interface and visualizations produced by the system. *U1* and *U2* indicate the two participants interacting with the respective systems. (a) Participant conversation: Property Extractor Architecture. (b) Participant conversation: Dialogue Manager Architecture.

- **RQ3 (*Usability*):** Do changes in the NLP engine affect users' perceptions of the NLI?

In this section, we explore the variables we considered in the analysis and the differences we found.

In order to answer the three research questions we posed in the introduction, we analyze the following aspects: the utterances processed by the NLI, the number of charts generated by the NLI after processing these utterances, types of charts generated and the charts selected by the users to aid their exploratory analysis. We also look at the database properties explored with the help of the CA given the same exploratory tasks, the conclusions drawn by the participants at the end of each task and lastly their responses to post study interview questions.

6.1 | RQ1: Functionality in Terms of User–System Interaction and Behavioral Influences

We analyze the user–system interaction in terms of utterances processed, number of charts generated and selected during the studies, and the types of charts that were generated and selected. For a deeper understanding of the user behavior during the tasks, we also examine the database properties explored with the help of the CA. Here, we describe the specifics of the analyses.

6.1.1 | Utterances Processed

In order to compare how the interactions with the CAs unfolded in the user study with the two systems—*PE* and *DM*, first we look

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TABLE 1 | Property Extractor Architecture: user interaction with the system.

Group	#Charts generated	% Charts selected	#Utterances	Average #words per utterance
1	35	82.86	429	7.67
2	109	15.60	538	6.66
3	158	15.19	557	9.67
4	78	29.49	731	6.51
5	27	48.15	663	7.44
6	47	57.45	683	6.01
7	84	53.57	474	15.37
8	90	27.78	417	4.99
9	148	40.54	538	9.34
10	122	19.67	536	7.52
Mean	89.98	39.03	556.50	8.12

TABLE 2 | Dialogue Manager Architecture: user interaction with the system.

Group	#Charts generated	% Charts selected	#Utterances	Average #words per utterance
1	48	13.16	152	10.59
2	37	47.50	163	8.36
3	35	37.14	128	10.97
4	36	41.67	161	8.56
5	24	68.00	235	8.76
6	34	55.88	143	5.90
7	37	68.42	180	6.79
8	48	41.18	240	10.06
9	49	28.00	293	8.14
10	34	61.76	188	7.74
11	62	37.50	311	7.91
12	51	51.92	222	7.55
13	15	73.33	165	6.10
14	28	28.57	201	8.68
15	43	59.09	314	6.13
Mean	38.73	47.54	206.40	8.15

at the utterances processed by each system. Note that the number of utterances processed are not the same as the number of utterances made by the participants. This is due to the processing latency of the utterances by the system. The average number of utterances processed per minute can be an indicator of responsiveness of the system while interacting with the users. Using the data in Tables 1 and 2, we find that on an average, PE processes 11.13 utterances per minute while DM processes 4.12 utterances per minute. Even though the mean of *#Utterances* for PE is greater than that of DM and *Average #Words per Utterance* is slightly higher for DM, we investigate if this difference, and the slightly higher *Average #Words per Utterance* in DM, are statistically significant. According to the Shapiro–Wilk test, only *#Utterances* has a normal distribution. Using Welch's T-test, we compare the means and find a significant difference (p-value < 0.0001) between the average number of utterances in the two-user study for PE and DM. For *Average #Words per Utterance* we conduct Mann–Whitney U test and do not find any significant association between the said factor and the user study sessions. Hence, we conclude that users talk significantly more in the PE version of the NLI with respect to the DM version. As discussed earlier in Section 4, DM has a more complex chain of processes, with several classification models in the NLU Module, followed by utterance-to-SQL conversion and finally retrieval of the data from the database. In PE, the whole process of retrieving data is done directly with the help of the Vega-Lite grammar. We speculate that once users realize that DM is relatively slow, they wait to obtain the resulting charts; this explains why they produce fewer utterances in DM. We will further discuss these findings in Section 7.

6.1.2 | Charts Generated and Selected

Next, we consider the number of charts generated and the percentage of generated charts that are selected during the studies.

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TABLE 3 | Property Extractor Architecture: types of charts generated and selected by the users.

Group	# Choropleth	% Choropleth sel.	# Heatmaps	% Heatmaps sel.	#Bar charts	% Bar charts sel.	# Line charts	% Line charts sel.	# Histograms	% Histogram sel.
1	26	88.5	0	0.0	1	100.0	7	71.4	1	0.0
2	14	50.0	0	0.0	33	27.3	58	1.7	4	0.0
3	89	23.6	2	0.0	9	0.0	58	5.2	0	0.0
4	12	66.7	12	83.3	12	16.7	42	7.1	0	0.0
5	18	50.0	2	100.0	0	0.0	7	14.3	0	0.0
6	21	85.7	0	0.0	10	50.0	16	25.0	0	0.0
7	57	56.1	1	100.0	5	60.0	20	40.0	1	100.0
8	13	76.9	10	40.0	18	22.2	48	14.6	1	0.0
9	87	51.7	1	100.0	12	58.3	48	14.6	0	0.0
10	26	26.9	0	0.0	25	12.0	60	11.7	11	63.6
Mean	36.30	57.61	2.80	42.33	12.50	34.65	36.40	20.56	1.80	16.36

TABLE 4 | Dialogue Manager Architecture: types of charts generated and selected by the users.

Group	# Choropleth	% Choropleth sel.	# Heatmaps	% Heatmaps sel.	#Bar charts	% Bar charts sel.	#Line charts	% Line charts sel.
1	33	15.1	2	0.0	13	0.0	0	0.0
2	32	46.8	1	100.0	2	100.0	2	50.0
3	25	44.0	6	83.3	1	25.0	0	0.0
4	30	36.6	5	60.0	0	0.0	1	100.0
5	20	75.0	1	0.0	1	50.0	1	100.0
6	19	47.3	0	0.0	2	40.0	10	80.0
7	30	63.3	5	100.0	2	100.0	0	0.0
8	38	42.1	4	50.0	3	60.0	1	0.0
9	39	28.2	5	40.0	1	20.0	0	0.0
10	25	52.0	5	100.0	3	75.0	0	0.0
11	41	36.5	0	0.0	4	57.14	14	35.7
12	41	48.7	4	50.0	5	83.3	0	0.0
13	14	71.4	0	0.0	1	100.0	0	0.0
14	23	30.4	3	33.3	0	0.0 0	0.0	
15	32	50.0	7	100.0	3	75.0	0	0.0
Mean	29.47	45.86	3.20	47.78	4.13	52.37	1.93	24.38

We also investigate the number of utterances by the two participants, the average number of words in an utterance, the types of charts (line charts, heatmaps, etc.) generated in the two study sessions and the percentage of each of those chart types selected by the participants. All these calculations are done by combining both tasks. In a preliminary evaluation of the PE version of our NLI [33, 34], we found that selection of generated charts—that is, clicking on the charts appearing on the conveyor belt in the interface to make it appear on the workspace, is a reasonable indicator of usefulness of charts. This motivated us to look into the percentage of charts selected for the two study sessions—overall number of charts selected by each group of PE and DM as well as for individual types of charts. At first glance, we find the average number of charts generated in DM is lower than in PE, whereas the average percentage of charts selected is higher in DM than in PE.

To assess the significance of this observation, we first check whether these factors have an underlying normal distribution and accordingly use a parametric or a nonparametric test. Using the Shapiro–Wilk test, we find both *#Charts Generated* and *%Charts Selected* to have normal distribution. We the use Welch's *T*-test to compare the means and among these two user interaction factors, only *#Charts Generated* is significantly different (p-value = 0.0061, < 0.05). This indicates that the difference in the number of did not occur by chance.

6.1.3 | Chart Types

Moving on to the types of charts generated, we look into the number of charts of each type (as discussed in Section 2) and the percentage of these charts selected by the users. Comparing Tables 3 and 4, we can see that PE produces more Choropleth maps than DM and *% Choropleth Map Sel*(*ected*) is also higher for PE. Similarly, PE produces more Line Charts but not many of them are selected by the users for their exploratory analysis. However, DM produces more Heatmaps and the selection percentage of this chart type is also higher than those for PE. However, despite PE generating more bar charts, the percentage of bar charts being selected by the participants is higher for DM. Additionally, while PE generated histograms during the user study, even if very few,

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the DM did not generate any at all, which explains why no histogram columns appear in Table 4.

We investigate whether any significant differences exist for these factors and start with tests of normality (Shapiro–Wilk test). Except *#Line Charts* and *% Line Charts Sel.*, the other four factors have normal distributions. After performing Welch's T-test to compare the means of the remaining four factors, we found no significant difference between the generation and selection of these chart types for PE and DM. For *#Line Charts* and *% Line Charts Sel.*, we conduct Mann–Whitney U test. We find that only for *#Line Charts* we can reject the null hypothesis (p-value < 0.0001), that is, there is a significant difference between the user study sessions, as concerns the number of Line Charts generated.

6.1.4 | Database Properties Explored During Tasks

Because these are exploratory tasks, and our interest is in how our NLI systems support such explorations, it is important to analyze whether the two systems engender different exploration behaviors in their users. Whereas a full analysis of temporal progression of the tasks is beyond the scope of this paper, insights into user behaviors can be gained by analyzing whether they explore different database properties. In what follows, we first illustrate the distributions of the properties explored in Figure 9a,b; we then apply χ^2 to see if the differences are significant; and finally, we perform post hoc analysis on the χ^2 results to assess which specific properties make the difference. As mentioned above, we start by looking at the percentages of each property in each task as shown in Figure 9a,b—within all the properties that were explored, the sum of Y-axis totals 100%. As we can see *COVID cases by date*, *access to doctor*, *African American population*, *poverty rate*, and *uninsured rate* have higher frequency in both tasks for PE. Whereas for DM, all other database properties except *African American population* have higher frequency for both tasks.

To find out if these observations are significant, we perform a χ^2 test of independence between the study sessions—PE and DM, and the database properties. We obtain $\chi^2 = 188.89$ for task 1 and $\chi^2 = 290.27$ for Task 2, p-value < 0.0001: in both cases, this shows significant association between the two variables. The strength of the association is checked using Crammer's V, $V = 0.4370$ for Task 1 and $V = 0.4396$ for Task 2. Using Cohen's interpretation of Crammer's V [45], we find medium association between the study sessions and database properties explored for both tasks. Further, we also perform a pairwise z-test post hoc analysis by looking at adjusted residuals for each cell of the contingency tables. Sharpe [46] and Agresti [47] discuss this post hoc analysis in their works where the largest differences between expected and observed counts, relative to sample size is calculated. Further to adjust for tests in multiple cells, we perform Bonferroni correction.

Significant associations between specific study sessions and database properties are reported in Table 5; the red color indicates a negative association (that property is discussed less than chance—for example, in Task 1, *county type* for PE); and blue, a positive association (that property is discussed more than chance—for example, in Task 1, *elderly percentage* for DM).

We observe that, for the first task there are fewer significant associations overall. Interestingly, the sign of the association switches between PE and DM for all three significant properties: *county type* and *elderly percentage* have a positive association with DM and negative with PE; this flips for *COVID cases by date*. Namely, users explore these specific three properties in "opposite" ways.

For the second task we observe more positive significant associations, especially for DM (eight positive associations), indicating an inclination of the users to explore the mentioned database properties more than expected by chance in DM. For PE we observe five positive association: as for Task 1, the positive association of PE with *COVID cases by date*, remains negative for DM.

6.2 | RQ2: Interpretability in Terms of User Inferences

At the end of each task, the participants of the study were asked to explain (as a group) the conclusions they had derived after exploring the data. The task questions were open-ended in nature; the participants were told that there were no correct answer and that they were free to draw conclusions from the observations made during the study. The participants explained their conclusions verbally to the study moderator; their explanations were audio recorded. These recordings were transcribed for analysis, which resulted in a conclusion text for each group in the PE and DM user study (as a reminder, each group provides task conclusions for each task). As mentioned in Section 1, the end-users should be able to understand the output of a system, a quality that makes a system interpretable (post hoc interpretability). Examining the conclusions drawn by users at the end of each task is a means of assessing the post hoc interpretability of the NLP engines. Thus, analyzing these conclusions help in understanding any differences caused by the two NLP engines when given the same set of tasks, thereby shedding light on any differences in the interpretability of the two systems. Here, we also leverage the open-ended nature of the tasks where the participants were allowed to freely explore the data. This allows for different conclusions to be drawn by each group, irrespective of whether it is the PE or DM user study. To fully explore these task conclusions, an in-depth manual annotation and analysis would be required, which would be very resource intensive. Rather, we started to gain insight into the potential differences in the conclusions of tasks in the user study, by looking at the semantic (dis)similarity in conclusions originating with dyads using the two different systems.

To get a snapshot of the key terms in the conclusions and for finding potential similarities and dissimilarities in the conclusions drawn by the participants of each user study we generate word-clouds using the textual transcripts of the conclusions. Word clouds provide a visual representation of a text where the frequency of a word is represented by its size and boldness in the cloud. Some of the high-frequency keywords in the conclusions of both tasks with PE and DM can be seen in Figure 10. Some of the most frequent words used by the participants for PE Task 1 (10a) are—*poverty*, *case*, *rate*, *rural*, *southeast*, *southwest*, *elderly*, *risk*; for DM Task 1 (10b), these are *rural*, *poverty*, *elderly*, *southeast*, *diabetes*, *access*, *doctors*, and *uninsured*. For Task 2, these are *case*, *poverty*, *rate*, *access*, *doctors*, *risk* for PE (10c) and *vulnerability*,

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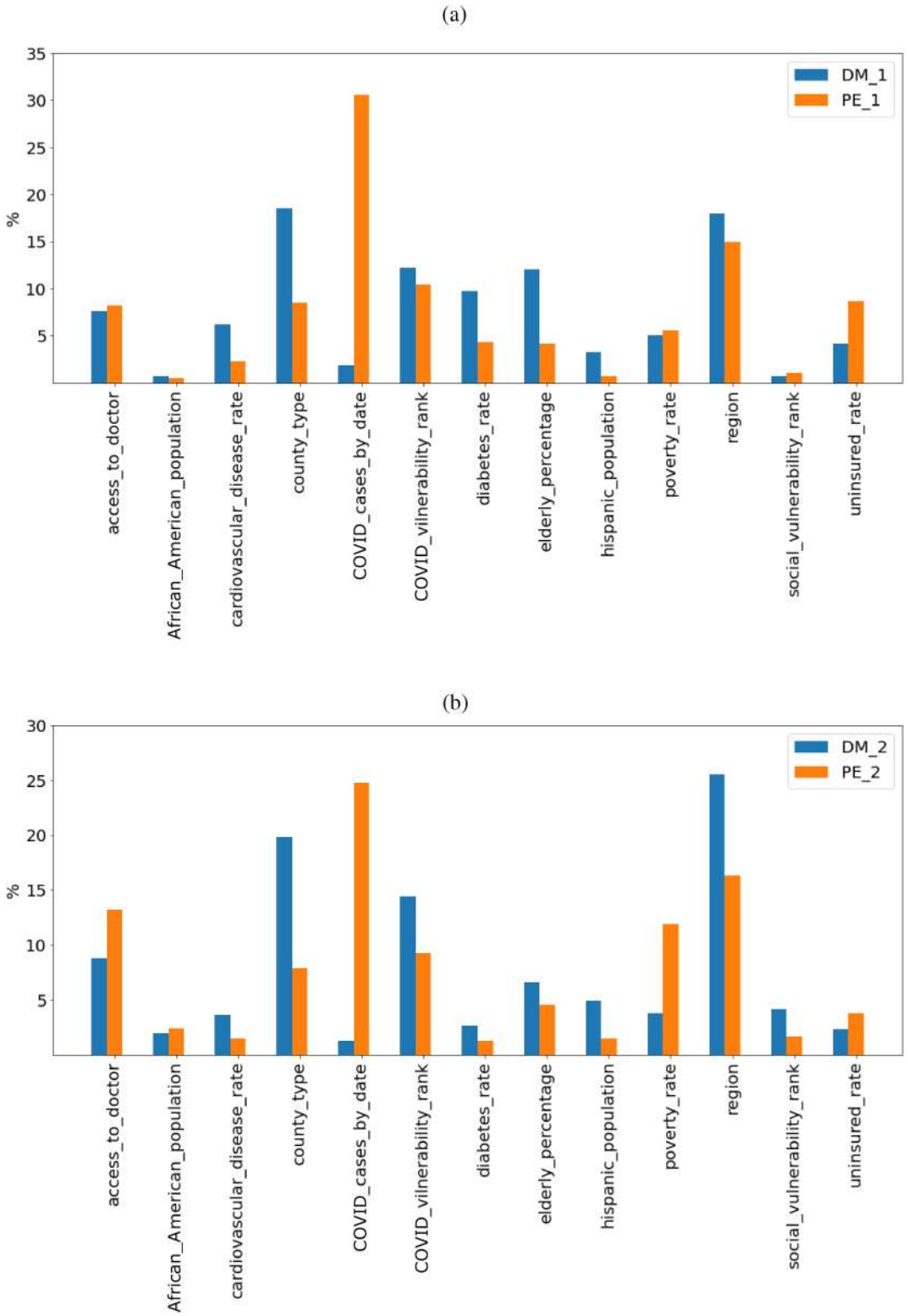


FIGURE 9 | Percentage of database properties explored. (a) Task 1: % of database properties explored in PE versus DM in Task 1; (b) Task 2: % of database properties explored in PE versus DM in Task 2.

rural, *diabetes*, *access*, *doctors*, *cardiovascular and disease* for DM (10d). These observations indicate some semantic differences in the overall theme in both the tasks for PE and DM.

In order to understand this semantic difference and make a comprehensive comparison, we measure the semantic similarity between the texts of the conclusions drawn in PE and DM user study. Semantic similarity quantifies similarity of meaning—low similarity is evidence for potential meaning divergence. We calculate the similarity of every conclusion from PE with those from DM (performing the analysis for Task 1 and Task 2 separately). First the conclusion text from each group is vectorized. This is

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TABLE 5 | Significant associations between user study sessions PE and DM with database properties explored during Tasks 1 and 2.

Database property	User study, Task 1
1. County type	PE
	DM
2. COVID cases by date	PE
	DM
3. Elderly percentage	PE
	DM
Database property	**User study, Task 2**
1. Access to doctor	PE
	DM
2. African American population	PE
	DM
3. County type	DM
4. COVID cases by date	PE
	DM
5. COVID vulnerability rank	DM
6. Diabetes rate	PE
7. Hispanic population	DM
8. Poverty rate	PE
	DM
9. Region	PE
	DM
10. Social vulnerability	DM

Note: Positive significant associations are color-coded with blue and negatives with red.

done by generating Sentence-BERT (SBERT) embeddings [48] for each sentence (from the transcripts) of these task conclusions. SBERT is a modification of BERT [49] for generating embeddings of sentences. SBERT embeddings help us represent each sentence in a vector space and the semantic similarity between two such SBERT embeddings can be calculated using a similarity metric such as cosine similarity, negative Manhattan distance, negative Euclidean distance, and so forth; here, we use cosine similarity.

In this analysis we calculate the semantic embeddings of the entire conclusion text of each group—these conclusion texts can have one or more sentences and SBERT can be used for generating embeddings of texts with multiple sentences. Thus, for Task 1, we have conclusions from 9 groups from PE (conclusion recording from one of the groups was not available due to technical issues) and 15 from DM making a combination of 135 similarity measures. Similarly, for Task 2, there are 150 combinations of similarity from 10 groups of PE and 15 groups of DM. We establish the cosine similarity threshold using the 75th percentile of the obtained scores. Thus, the threshold for Task 1 is 0.64 and for Task 2 is 0.59. Therefore, conclusion pairs having similarity score above these thresholds for the respective tasks are considered to have a semantic overlap for this analysis. In Figures 11 and 12, we see examples of conclusions by participants for Task 1 and Task 2, respectively. In both these examples, we compare conclusions from one of the groups in PE with two from DM, one having cosine similarity score higher than the threshold and the other lower than the threshold. Overall, we see a 26% semantic overlap for both the tasks based on the respective thresholds—namely, no matter the task, only 26% of conclusion pairs (one from a PE group, one from a DM group), have a cosine similarity higher than the threshold. Kindly note that the % of semantic overlap presented here is for the overall comparison of the two systems by combining Tasks 1 and 2.

6.3 | RQ3: Usability in Terms of User Perceptions

After finishing both tasks in the study, the participants were asked to individually respond to a semi-structured interview; the interview contained questions about demographics (*what's your age*), previous experience with visualization tools, and experience with the NLI (e.g., *Please describe any features that you would like to see in the visualization application*). Additionally, it contained two questions on a 5-point Likert scale, *How useful were the generated charts* and *Rate the command system*. Here, we discuss the responses of the participants from both PE and DM to these two questions (given their nature, the other questions were meant more as food for thought for our future work than as metrics to compare the two NLIs):

1. *How useful were the generated charts on a 5-point Likert scale*: Subsequently for ease of use, this question is referred to as *Usefulness Rating*. *Usefulness*: Figure 13a, a diverging stacked bar chart is used to represent the Likert scale ratings provided by the users. We use this chart type based on the recommendation of Robbins and Heiberger [50] and Heiberger and Robbins [51] that diverging stacked bar charts are the best visualizations for Likert scale data. On the vertical axis, it shows the user responses to the question for both PE and DM. The dashed vertical line shows the divergent point running through the bar for *Neutral* rating. To its left are the stacked bars representing the negative ratings *Not Very Useful* and *Somewhat Useful*; and the positive ratings *Useful* and *Very Useful* to its right. Thus, the horizontal axis shows the absolute count of negative and half of the neutral ratings to the left of the divergent point, and the absolute count of half of neutral and the positive ratings to the right. The numbers on the stacked bars show the absolute counts of the responses belonging to the respective category. Thus, from the provided ratings, respectively, 80% and 66.6% of users found the charts generated by the PE and DM NLIs *Useful* or *Very Useful*. To get a deeper understanding of these ratings and to find any significant difference between the ratings provided for PE and DM we perform Mann–Whitney U test and find that there is no significant statistical difference. We obtain a median of 4 for both study sessions; Mood's Median Test confirms there is no statistical difference.
2. *Rate the command system on a 5-point Likert scale*: We refer to this question as *Ease of System Use Rating* in what

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FIGURE 10 | Keywords from task conclusions. (a) PE Task 1. (b) DM Task 1. (c) PE Task 2. (d) DM Task 2.

follows. *Ease of System Use*: Figure 13b shows the diverging stacked bar chart for the users' Likert chart responses to this question (reads similar to the one for *Usefulness Rating* described earlier). Based on these ratings, respectively, 35% and 43.3% of users found the command system of PE and DM *Easy* or *Very Easy* to use. Further, we also perform Mann–Whitney *U* test and find no statistically significant difference between groups PE and DM. Mood's Median Test also confirms no significant difference, the median rating for both being 3.

7 | Discussion and Takeaways

In this paper we systematically analyze the findings of the user study to answer the three research questions we posed at the beginning:

- **RQ1 (*Functionality*):** Do changes in the NLP engine affect user behavior, and if yes, how?
- **RQ2 (*Interpretability*):** Do changes in the NLP engine affect the conclusions that users draw from their explorations, and if yes, how?
- **RQ3 (*Usability*):** Do changes in the NLP engine affect users' perceptions of the NLI?

In this section, we present the results obtained from the systematic analysis of each of the above research questions. Following this, we summarize the key takeaways from these analyses.

7.1 | RQ1: Functionality, User–System Interaction and Behavioral Influences

The findings described in Sections 6.1.2 and 6.1.3 show that there is a statistically significant difference between PE and DM as concerns the *#Utterances* processed, *#Charts Generated*, and *#Line Charts* generated, among all the other factors considered for the analysis. We pointed out that DM is slower at processing utterances compared to PE, contributing to the significant difference in *#Utterances*. The difference in *#Utterances* affects *#Charts Generated*, since both of these factors are consequences of the user–system interaction. Regarding *#Line Charts*, significantly more are generated with the PE NLI. Line charts are generated during the visualization of the timeline of some event, which in this scenario pertains to any visualization related to the number of *COVID cases by date*.

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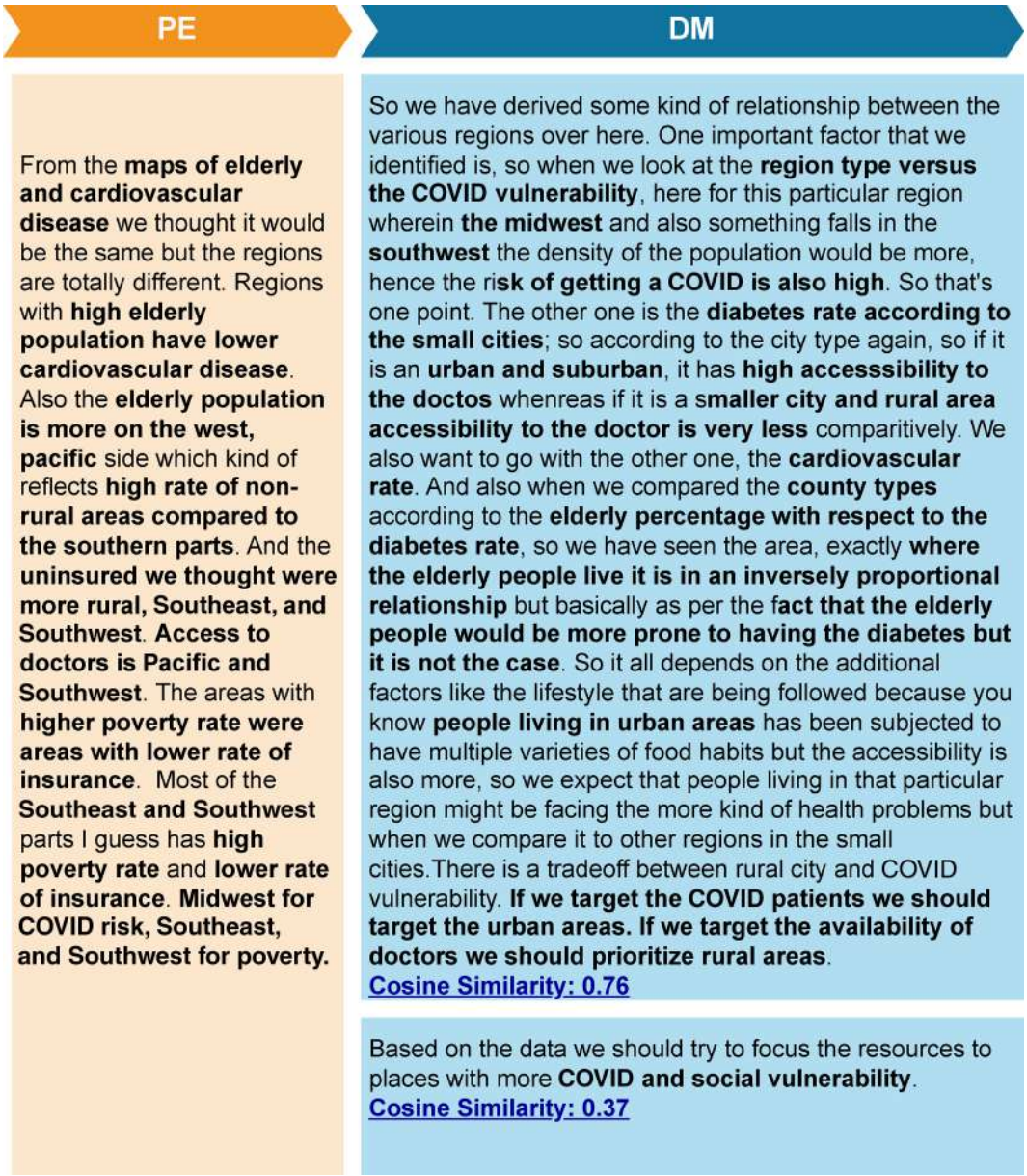


FIGURE 11 | Examples comparing conclusions for Task 1.

While investigating database properties explored during the tasks, we see for both Task 1 and 2, there is a positive association between PE and *COVID cases by date*. Again, when we look at the task conclusions we also find the word *Case* has high frequency for both tasks of PE. Both of these indicate that participants discussed more about *COVID cases by date* thus generating significantly more *#Line Charts* and while drawing conclusions of the exploratory tasks, they talk more about the number of COVID cases.

Indeed, the inspection of significant database properties explored in the two tasks reveals that PE has a significant positive association with *COVID Cases by Date* for both tasks 1 and 2. As discussed earlier, this can be both a cause and an effect of the higher number of Line Charts generated, in a kind of reinforcing feedback loop—because of PE's proactive nature in presenting charts that were not requested, and its property-oriented recognition, most spontaneous charts are line charts about *COVID Cases by Date*. In turn, this behavior may affect the participants who, seeing these charts pop up, keep refocusing on this specific feature.

During Task 2, it is DM that has more significant positive associations (8) with database properties compared to PE (4) (excluding *COVID cases by date*, as shown in Table 5). We speculate that the slower DM may give users more time to think about what to explore next; on the other hand, PE which provides them with more charts including the spontaneously generated ones, may drive them to focus on some specific properties only.

7.2 | RQ2: Interpretability and User Inferences

The semantic overlap in the task conclusions gives us an idea of the extent to which the take-aways of the two tasks vary in the two systems PE and DM—potentially a different interpretation of the same overall architecture, the same set of tasks when the

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FIGURE 12 | Examples comparing conclusions for Task 2.

internal NLP engine is different. It also gives us a picture of the "topics" that were explored by the participants in the user study and whether differences by conditions exist. This suggests that the underlying architecture of such data exploration systems may have a direct impact on the outcomes of the exploratory tasks. The differences in properties explored is reflected in the conclusions drawn by the participants at the end of tasks. Some of the common "topics" include emphasis on the *Southeast* and *Southwestern* regions of the United States, *Rural* county type and *Poverty rate* for both tasks. While there is much focus on data exploration based mainly on number of COVID-19 cases in various counties in the United States in PE, we observe that participants in the DM studies speculate about other factors from the provided data like *diabetes risk*, *cardiovascular diseases*, and so forth. The findings from this comparison suggest that, given the same two tasks, the conclusions drawn overall by participants in the PE and DM user study are semantically different. This ties back to our hypothesis that a system with a specific tendency to generate certain types of data visualization may affect the conclusions drawn by the users, even more so since exploratory data analysis is open ended in nature.

7.3 | RQ3: Usability and User Perceptions

According to the participants' responses to two post-study interview questions, there is no significant difference as concerns the user experience between the two systems. In general, for both NLIs, users like working with them and find them useful—combining the results from Section 6.3, on an average 72% of users find the charts generated by the NLI useful (rated as *Useful* or *Very Useful*). This confirms previous studies including our own background studies [33, 34, 38], as well as other related works described in Section 2.

Likewise, there is no significant difference between the two systems as concerns ease of use. In general, when we combine the ratings from the two studies from Section 6.3, on an average 40% of participants find the NLIs *Easy* or *Very Easy* to use, with another 22% choosing the neutral point on the scale. finding them neither *Easy* not *Difficult* Our hypothesis is that, since the participants were not given any specific instruction for interacting with the NLI, they had to learn to use the NLI for performing the exploratory tasks on the fly: this must have been challenging

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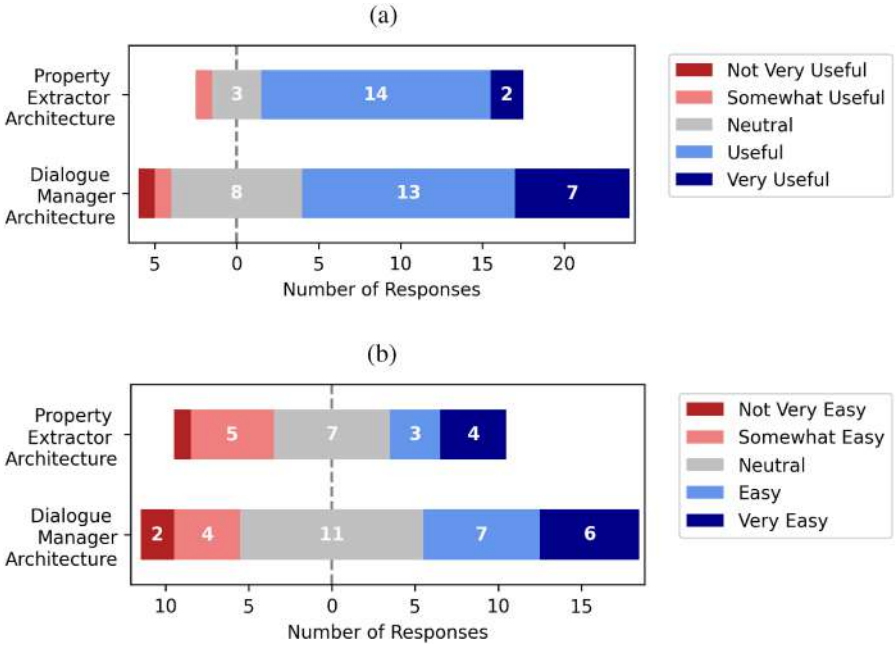


FIGURE 13 | User response to post study interview questions in 5-point Likert scale. (a) Usefulness of generated charts rating. (b) Ease of system use rating.

and cognitively demanding since their main focus was on collaborating with their study partner to perform the two given tasks. However, we see on an average between PE and DM, only 25% participants across the user study found the command system *Not Very Easy* or *Somewhat Easy* to use.

A disclaimer: Our speculations as to why the two systems would lead the users to different conclusions need to be further substantiated. One way to do so would be an extremely time consuming analysis that goes back to the video recordings and codes them for pertinent variables, for example: do users suddenly change the subject (to assess whether PE truly distracts them)? is it while waiting for DM to answer that users of DM explore additional properties?

7.4 | Takeaways

To the best of our knowledge, we are the first to conduct user study specifically focusing on a CA to support the democratization of data visualization, revealing important insights into the functionality, interpretability, and usability of such systems. From the comparative analysis of the two systems we can conclude that an ideal CA for the purpose of exploratory analysis should be the one in which there is some latency in generation of the visualizations so that users are not overwhelmed by the system. Our hypothesis is that this will help users make meaningful inferences from data they are exploring. Next, our findings show that user–system interaction is moderately influenced by the type of NLP engine used, which is crucial for improving engagement and in-turn system perception [52]. Next, the interpretability of the CA is significantly impacted by the choice of NLP engine, affecting users' ability to understand and trust the system's outputs [53]. Despite these variations, the overall usability of the system remains mostly unaffected by changes in the NLP engine. This suggests that users can maintain efficiency across different NLP engine implementations. These findings are crucial given the current popularity of artificial intelligence(AI) and LLMs which are being commonly used as black boxes in various applications [54]. This work demonstrates the potential of different NLP engines to shape user perceptions and interactions with task-oriented systems, emphasizing the need for transparent and user-friendly AI designs.

8 | Future Work

We propose several venues of expansion for the work we presented in this paper. We start with extensions to the NLIs in their current form, or suggest further analyses for the user study with two NLU engines we have run; we then move to broader implications of this type of work, and potential venues of research in this larger space.

8.1 | Extensions to Current Work

The first two extensions pertain to the fact that the users used speech, but the NLIs only made use of potentially wrong word transcriptions. We realized that one bottleneck to a smooth interaction between users and NLI in our studies was caused by the errors due to speech to text transcription. We plan to utilize more advanced speech to text algorithms like Whisper API[13] which were unavailable when our NLIs were deployed (summer and fall 2022). Additionally, human speech is not just words; it is characterized by many features, from phonetic (articulation, stress, pitch), to prosodic (intonation, tune) to interactional (fillers like "uhmm," repetitions), and so on. We aim to compare the robustness of the two NLP engines by quantifying the correctness of the generated charts given these features of the spoken interaction.

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Second, human interaction is characterized by multimodality: nonverbal components are crucial in human communication. We believe that leveraging nonverbal cues like hand gesture, eye gaze and head movements can go a long way in improving the overall experience with CAs. One of our earlier prototypes [30, 55] was in fact able to interpret pointing gestures to the large screen; however, the setting was designed for a single user. Once we moved to two collaborators, it became too complex to adapt our previous approach to gesture recognition: we then reverted to the users using a mouse, as described in Section 5. We intend to further explore additional modalities in future work.

Third, as concerns further analysis of the user study we conducted, as we noted, some of the effects we observed deserve further investigation if resources allow. For example, we speculated that the different speeds with which the NLIs respond, may explain the fact that the users interacting with DM especially in Task 2, explore more properties than the users interacting with PE. To further investigate this potential explanation, we could annotate the video recordings, perhaps in a semi-automatic fashion, for *lines of reasoning* (*LOR*) [56, 57]. LORs have been used in various fields, including education, to model *an argument or set of arguments leading to a conclusion* [56]. LORs would highlight where participants start talking about one property and make inferences about it, whether they keep talking about it or switch to another one, and more importantly, how a new chart either makes them continue the same LOR, smoothly leads them to starting another LOR, or rather distracts them causing an abrupt conclusion to the original LOR and the start of a new one.

Likewise, insights could be derived by further analysis of the transcribed conclusions that the users provided. So far, we have pursued rather simple word-based analyses, namely, word embeddings and coarse measures (cosine similarity, word clouds). Annotating LORs would also be appropriate in this case, to analyze not just which properties users mention in their conclusions, but how those properties support their conclusions.

Manual annotations of LORs is prone to be extremely time consuming, but even semi-automatic annotation is beyond the state of the art: in fact, extracting LORs automatically is an unsolved problem which has not received as much attention in the literature [58, 59], although a recent paper does use BERT to do so [60].

8.2 | Broader Implications and Potential Extensions

We now turn to some broader considerations, which are interconnected: trustability, transforming the NLI into a true CA, and the role of LLMs in the whole enterprise.

As we discussed in Section 6, one finding of our analysis is the fact that the two NLIs appear to have led their users in somewhat different directions, with users of DM exploring more database properties. We also highlighted that the conclusions that the users reached with the two NLIs somewhat diverge, at least when measured at the word level, via word embeddings and cosine similarity. A larger implication of this analysis is that these technologies can be supportive of exploration and hypotheses generation, but at this point in time, they appear unable to help participants reach correct conclusions, because as shown users reach divergent conclusions without having a sense of how valid they may be. Namely, these technologies can greatly enhance exploration, but they can potentially (and inadvertently) generate great misinformation and confusion. Our study so far, does not show which NLI (if any in fact) supports users in deriving conclusions which are closer to valid explanations for the patterns observed in the data, and whether these technologies can effectively increase the quality of decision-making.

To truly trust the NLI, and conversely, to make sure that the decision-making is truly grounded in reality, we believe that we must move away from NLIs that can only produce charts, to NLIs that can actually participate in conversation and reasoning. For example, the NLI should be able to provide explanations to users as to why it produced the charts it did; it should be able to ask clarification questions in case of uncertainty as concerns understanding the user input; and it should be able to suggest properties to explore, and more in general, elucidate potential connections and patterns, spontaneously or on request, precisely like a knowledgeable human collaborator would.

We conclude by speculating on the role that LLMs may play in further development. In Section 2, we discussed Chat2Viz [22], which explored using three different LLMs to generate visualizations from NL with promising results; but we also observed that this work does not take place in a conversational setting, and was evaluated on the nvBench dataset, not with real users. And in contrast with these results, in some preliminary experiments we recently ran with ChatGPT applied to our conversational requests, we observed it is able to produce only the most basic chart types.[14] The usage of LLMs also directly affects trustability as we discussed above; the popular press is full of examples of LLM's "hallucinations": their opaque nature makes the issues we discussed earlier regarding users reaching valid conclusions, even more concerning. We do not yet see a clear path forward on how to use LLMs in this enterprise as proficient and validated participants; but it is clear that this is an important area of research to explore. A potential usage is as generators of actual utterances that an advanced conversational NLI should produce, namely, as the last module. In today's day and age of LLMs and popularity of AI applications like ChatGPT, we plan to experiment in this direction. However, prompt engineering is a main aspect of suh models. Since our goal is to faclitate a more natural interaction with the CA, we believe this will be an interesting future direction.

Author Contributions

Abari Bhattacharya: contributed to the idea, formulated the research question, developed the methodology, conducted the user study, performed data analysis, and wrote the manuscript. **Barbara Di Eugenio:** co-principal investigator (co-PI) of the project, contributed to the idea, formulated the research question, and wrote the manuscript. **Veronica Grosso:** contributed to the idea, testing the methodology, and conducted the user study. **Andrew Johnson:** principal investigator (PI) of the project, contributed to the idea, formulated the research question, feedback on the manuscript. **Roderick Tabalba:** contributed to the idea, development of the user interface, and user study. **Jason Leigh:** principal investigator (PI) of the project, contributed to the idea, user study,

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and provided feedback on the manuscript. **Moira Zellner:** co-principal investigator (co-PI) of the project, contributed to the idea, user study, and provided feedback on the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ENDNOTES

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[3] We acknowledge that this task may be fraught in the era of Black Lives Matter in the United States. These data were collected prior to 2020, when the current awakening as concerns policing and racism surfaced to public consciousness.

[4] The transcribed and annotated corpus, including the visualizations, is available at https://github.com/uic-nlp-lab/Chicago-crime-vis-corpus. At this same address, an augmented dataset is also available, built using delexicalization and paraphrasing, which increases the size of the original data 10 times.

[5] https://vega.github.io/vega-lite/.

[6] https://wicg.github.io/speech-api/.

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[12] Also available at: https://github.com/Tabalbar/articulate-plus under the folder "Supplementary Material."

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